



**Laboratory Manual
for
SC1003
Introduction to Computational Thinking and
Programming**

**Practical Exercise #6:
Procedural Abstraction**

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Ex. #6 – Procedural Abstraction

Learning Objectives

Understand and implement procedural abstraction in Python. Focused on function development and function call.

Equipment and accessories required

PC/notebook with python compiler (eg python IDLE)

1. Procedural abstraction

Procedural abstraction is a fundamental concept in computer science and software engineering that involves the separation of the "what" from the "how" in the context of functions or procedures. It allows programmers to define a sequence of operations (a procedure) without specifying the details of how those operations are implemented. This abstraction helps in managing complexity by breaking down a program into smaller, more manageable pieces.

In this exercise, you will learn how to encapsulate specific tasks or operations, allowing you to group related code together. This makes the code easier to read, maintain, and debug.

In this exercise, you will continue to work on the code developed in Tutorial 6 (procedural abstraction) for discussion 3 & discussion 4.

The code was enhanced to demonstrate the procedural abstraction via function development for the context of the 1A2B (Bulls & Cows) game. The main components include the following procedure and the details are specified at the sample code below:

1. **Randomly generate the target number:**
2. **Obtain the guessing from users:**
3. **Compare the user guessing with the target:**

If you could not design your prototype of function in discussion 2, then copy the following hint code to your IDE, e.g., IDLE, follow the TODO task lists and sample output as follows to complete the exercises.

```
import random

WIDTH = 4
numList = [1,2,3,4,5,6,7,8,9,0]

#####
# functionality: randomly generate the target number
# input:
#   numList: the list with 10 digits
#   count: the number of digits to generate
# output: a list with four digits (integers)

def genNumbers(numList, count):

    ## TODO ##
    ## Shuffle the numList and return the shuffled list

    return numbers
#####

#####
# functionality: obtain the guessing from users
```

Ex. #6 – Procedural Abstraction

```
# input: None
# output: a list with four digits (integers)

def userGuess():

    inputStr = input("Please input 4 digits: ")

    ## TODO ##
    ## Check user input and ensure user entered correct
    ## datatype and length of input.

    guess = []

    ## TODO ##
    ## Transfer the user input to guess list.

    return guess
#####

#####
# functionality: compare the user guessing with the target
# input:
#   guessList: the list of four digits (user guessing)
#   answerList: the list of four digits (target)
# output: a tuple (#A, #B)

def checkGuess(guessList, answerList):
    ## TODO ##
    ## Check how many correct number entered by the user
    ## and return the count (bulls, cows).

    return (bulls, cows)
#####

#####
# Main program putting all function together

answer = genNumbers(numList, WIDTH)
#print(answer)          ## UNCOMMENT THIS TO SEE WHAT WAS THE
                        ## SHUFFLED LIST

attempts = 0

while True:
    guesses = userGuess()
    attempts += 1
    #print(guesses)

    result = checkGuess(guesses, answer)
    #print(result)

    if result[0] == 4:
        print("You Win!!")
        print("Attempts: ", attempts)
        break
    else:
        print(result[0], "A (Bulls)", result[1], "B (Cows)")
```

Ex. #6 – Procedural Abstraction

Sample output:

Please input 4 digits: 1234
0 A (Bulls) 1 B (Cows)
Please input 4 digits: 2123
0 A (Bulls) 1 B (Cows)
Please input 4 digits: 222
Wrong format!! Please input 4 digits: 3213
1 A (Bulls) 1 B (Cows)
Please input 4 digits: 3214
1 A (Bulls) 0 B (Cows)
Please input 4 digits: 1928
0 A (Bulls) 1 B (Cows)
Please input 4 digits: 8732
1 A (Bulls) 2 B (Cows)
Please input 4 digits: 2323
0 A (Bulls) 2 B (Cows)
Please input 4 digits: 23
Wrong format!! Please input 4 digits: 2323
1 A (Bulls) 1 B (Cows)
Please input 4 digits: 3
Wrong format!! Please input 4 digits: 3333
1 A (Bulls) 3 B (Cows)
Please input 4 digits: 3569
2 A (Bulls) 2 B (Cows)
Please input 4 digits: 3596
You Win!!
Attempts: 15